

Research papers

Conjunctive optimal design of water and power networks

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ABSTRACT

Water distribution systems (WDS) and power grids (PG) are critical infrastructure systems that are vital to all human activity. As such, their quality of service is of great importance for economic, environmental, and human welfare reasons. Although traditionally being analyzed separately, the two systems are interconnected and can mutually affect one another. In order to utilize the potential benefits that the two systems can produce for each other, their design and operation should be analyzed conjunctively. In this paper, a conjunctive optimal design approach for water and power networks is presented, with the objective of finding the dimensions of the systems' facilities that will result in minimal overall costs, for both design and operation. The model is formulated and implemented on two example applications using an off-the-shelf nonlinear solver by MATLAB and compared to the optimal design of the independent WDS. A sensitivity analysis is performed to provide validity to the obtained results. The conjunctive design is compared to the design of an independent WDS to emphasize the effect of including the PG in the optimization problem. Results show a clear link between the availability of renewable energy and sizing of WDS components. The design of the independent WDS leads to the violation of PG constraints, which are satisfied when including both systems under a single optimization model, demonstrating the importance of a holistic design approach.

1. Introduction

Water distribution systems (WDSs) are a collection of hydraulic elements such as pumps, pipes, tanks, and valves, designed to convey water from sources to consumers. The behavior of the system is determined by physical laws of mass and energy conservation; water demands; and its layout. The optimal design problem of WDS is the task of sizing certain elements in the system such that would satisfy its constraints, such as the supply of nodal demand, minimal pressure, and water quality standards, and result in an optimal objective value, traditionally set as minimal total construction and operation cost (Perelman and Ostfeld 2007).

Designing an optimal water distribution network presents a multifaceted challenge that requires careful consideration of hydraulic performance, cost efficiency, and sustainability. Engineers must balance the complex interactions between pipe diameters, network topology, and operational constraints to ensure reliable water supply and efficient system operation. Factors such as pressure requirements, flow rates, water quality standards, and resilience to external disruptions further complicate the design process. Optimization involves minimizing both

capital and operational costs while maximizing system reliability and resilience to future demands and uncertainties. The variables that are considered in such optimal design problems are traditionally pipe diameters, tank volumes and pumping station capacities.

Due to the non-linear and non-convex relationship between flow and head in WDS, and the use of discrete diameter sizes, the problem is regarded as highly complex (Bragalli et al., 2011) and found to be NP-Hard (Yates et al., 1984). Linear programming techniques can be utilized (Alperovits and Shamir 1977) to minimize costs while satisfying constraints on pipe diameters, flow rates, and pressure requirements. During the 1980's and 1990's several non-linear solution methods have been developed such as enumerations (Gessler 1985) and decomposition (Fujiwara and Khang 1990, Eiger et al. 1994). Over the past three decades, a significant amount of research has concentrated on the development and application of heuristic methods to solve the optimal design problem. Those methods can allow the efficient exploration of large solution spaces, which are tackled as part of the optimal design problem. Among those methods are genetic algorithms (Savić and Walters, 1997), simulated annealing (Zoppou and Reed 2001), shuffled from leap algorithm (Eusuff and Lansey 2003), differential evolution algorithm

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(Suribabu 2009), particle swarm optimization (Montalvo et al. 2008), ant colony optimization (Ostfeld and Tubaltzev 2008) and whale optimization algorithm (Ezzeldin and Djebdjian 2020). Several examples also exist for the combination of analytical methods with heuristic ones (Zheng et al. 2014, Reza et al. 2017).

Power grids (PGs) are networks of transmission lines, substations, transformers, and distribution points designed to deliver electricity from power plants to consumers. Designing an optimal power grid is a complex challenge that requires balancing numerous factors to ensure efficiency, reliability, and sustainability. Engineers must consider geographical factors such as terrain and population distribution, as well as economic variables like cost-effectiveness and scalability. The grid must be robust enough to handle peak demands while also integrating renewable energy sources effectively to reduce environmental impact. Furthermore, technological advancements in energy storage and transmission play a crucial role in enhancing grid resilience and stability. Achieving an optimal design involves navigating these multifaceted considerations to create a system that meets current and future energy needs efficiently and sustainably. Traditionally, decision variables for the power grid optimal design problem are generator capacities, sizes of energy storage facilities (Yang et al. 2020), transmission line sizes (Khalil and Ismail 2006), and the integration of renewable energy sources (Garni et al. 2018; Majidi et al. 2022). A similar problem which is often tackled in research is the problem of power grid expansion, dealing with the planning of an additional section of the grid (Kim and Hur 2021, You et al. 2016). These problems are deemed mathematically complex, due to the non-linear and non-convex nature of the power flow equations (Lavaei and Low 2010). Common methods used to overcome the computational challenges are linear and quadratic programming (Fortenbacher and Demiray 2019), second-order cone programming (Baradar et al. 2013), semidefinite programming (Molzahn et al. 2013), along with many other relaxation and approximation methods (Molzahn and Hiskens 2019).

The link between the systems has been studied from various perspectives (Hamiche et al. 2016), with most work linking WDSs and PGs concerned with their joint operation (Oikonomou and Parvania 2019; Takahashi et al. 2017; Sun et al. 2018). Hari et al. (2023) have considered a joint expansion problem, designing storage and generation facilities for the conjunctive system. In their work, they define the objective as maximizing the demand that is supplied by the system, rather than the traditional minimal cost objective. However, the network considered there is quite simplistic and does not consider loops, multiple water consumers, or multiple storage and pumping facilities, which can significantly complexify the problem. Yao et al. (2023) proposed an optimal design problem of WDS under uncertainty, combining the uncertainty in water demand with uncertainty of solar power availability. A behind-the-meter (BTM) component was also designed as part of the work to support the supply of power to the WDS. The PG itself and its constraints were not considered. They found that while robustness of the solution is improved, the design of a solar component did not affect the sizing of components inside the WDS. Yao et al. (2021) suggested to optimally design water tanks in the system to enhance the flexibility of the power system and to improve water quality. In their work they do consider the PG's structure and constraints, but the problem focuses on tanks rather than the design of the entire system, and without any decision variables on the PG side.

This paper proposes a conjunctive optimal design problem, combining the design of a WDS and a PG to explore ways in which the sizing of components in one system affects the other and the operational policies of both systems. On the WDS side, capacities of storage and pumping facilities are determined, and on the PG side – the capacity of a solar field is determined. This paper is aimed to provide two main contributions to existing literature. The first is the very presentation and mathematical formulation of the conjunctive design problem, which represents a relatively novel and more holistic approach to water resources management, taking into consideration the explicit modelling of

adjacent and interconnected systems (in this case, a power system), as opposed to the traditional approach which usually considers merely the WDS standpoint. Hopefully, this will spark a discussion about the joint examination of those two interconnected infrastructure systems. The second contribution lies in the analysis of the results, which clearly shows how those two systems depend on and even constrain one another. The joint analysis provides unique and often counterintuitive insights into the link between different components in those systems and proposes ways to design them together in a way that takes advantage of those links.

The rest of the paper is organized as follows: section 2 includes a mathematical formulation of the conjunctive design problem; section 3 details the solution of two example applications, including sensitivity analyses; and section 4 includes the conclusions of the work. The work has the potential to help design better infrastructure in the future, which in turn will increase the systems' efficiency, reduce operational costs, and environmental impact.

2. Problem formulation

The formulation of the conjunctive design problem is brought here, comprising all constraints and decision variables that form each of the two problems of WDS optimal design and PG optimal design.

2.1. Decision variables

Since a system's operation is highly dependent on its design, the operational costs must be considered as part of the problem's objective (as will be discussed further in the Objective function sub-section). Hence, the decision variables include also operational decisions and could be divided into two categories – design variables, and operational variables.

The design decision variables for the conjunctive design problem are: P_{max} [kW] – pumping capacities of all pumping stations in the WDS; V_s [m³] – storage capacities of all tanks in the WDS; SF [MW] – solar field capacities of all solar fields to be constructed and connected to the PG. The operational decision variables are: Q [m³/hr] – flows in all links in the WDS; H [m] – heads in all nodes in the WDS; n [rpm] – pump rotation speeds in all pumps in the WDS; p_{in} [p.u], q_{in} [p.u] – active and reactive power generation quantities at all generators in the PG; p [p.u], q [p.u] – active and reactive power at all buses in the PG, respectively; v [p.u] – voltages at all buses of the PG. As opposed to the design decision variables, the operational decision variables are to be determined for each timestep of the optimization horizon.

Fig. 1 shows the water and power demand patterns that were used, along with a solar generation curve that limits the amount of power that could be generated at any given hour of the day (Jahid et al. 2018).

2.2. Constraints

The optimal design problem includes the following constraints on the WDS side, which constitute node mass conservation (1), tank continuity (2), tank closure (3), pump curve (4), link energy conservation (5), node head bounds (6), pump rotation speed bounds (7), and non-negative pump flow (8), and head (9):

$$A^T \cdot Q^t = d^t \quad \forall t \in \mathcal{T} \quad (1)$$

$$H_{tank,s}^t = H_{tank,s}^{t-1} + \frac{Q_{tank,s}^t \cdot \Delta t}{\alpha_{tank,s}} \quad \forall t \in \mathcal{T} \forall s \in S \quad (2)$$

$$H_{tank,s}^{24} \geq 0.95 \cdot H_{tank,s}^0 \quad \forall s \in S \quad (3)$$

$$H_{p,l}^t = a_{pumps,l} \cdot \left(\frac{n_{p,l}^t}{n_{max}} \right)^2 - b_{pumps,l} \cdot (Q_{p,l}^t)^2 \quad \forall l \in L \forall t \in \mathcal{T} \quad (4)$$

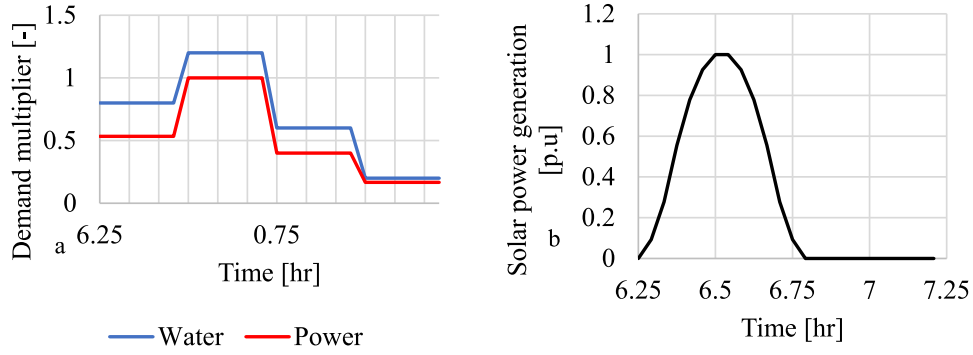


Fig. 1. Demand patterns (a) and solar profile for solar generator (b) (Jahid et al. 2018).

$$A \bullet H^t = -RQ^t|Q^t|^{0.852} - A_0 \bullet H_0 + A_{pumps} \bullet H_p^t \quad \forall t \in \mathcal{T} \quad (5)$$

$$H_{min}^t \leq H^t \leq H_{max}^t \quad \forall t \in \mathcal{T} \quad (6)$$

$$n_{min}^t \leq n_{p,l}^t \leq n_{max}^t \quad \forall t \in \mathcal{T} \forall l \in L \quad (7)$$

$$Q_{p,l}^t \geq 0 \quad \forall t \in \mathcal{T} \forall l \in L \quad (8)$$

$$H_{p,l}^t \geq 0 \quad \forall t \in \mathcal{T} \forall l \in L \quad (9)$$

where: A^T is the network's incidence matrix; d is vector of nodal demands; $H_{tank,s}$ is the total head of water in tank s [m]; $Q_{tank,s}$ is the flow of water in the pipe connected to tank s [m³/hr]; Δt is the length of a timestep [hr]; $\alpha_{tank,s}$ is the cross-section area of tank s [m²]; $H_{tank,s}^0$ is a given initial condition for the head of tank s ; $H_{tank,s}^{24}$ is the elevation of tank s at the end of the simulation period [m]; R is a column vector of pipe resistances; A_0 is a matrix with elements {0,1}, which defines the columns corresponding to the sources in the graph matrix; H_0 is a column vector containing the values of constant heads at the sources [m]; A_{pumps} is a matrix containing pump links indicators inside the WDS, used to adjust the dimensions of matrix H_p^t ; $\bullet$ is the Hadamard product (elementwise multiplication); $a_{pumps,l}$ and $b_{pumps,l}$ are the curve coefficients of pump l ; $n_{p,l}^t$ is the pump speed of pump l ; n_{max} is the maximum pump speed [rpm]; H_{min} and H_{max} are vectors of the upper and lower head bounds at the nodes [m]; and n_{min} and n_{max} are the minimum and maximum pump rotation speeds, respectively.

This formulation is novel in the sense that it allows formulating the problem as a non-linear program (NLP), instead of the well-known mixed-integer non-linear program (MINLP) formulation, as it is usually modelled in other studies found in the literature. The optimal operation problem of WDS is categorized as an MINLP, as it required the use of integer variables to correctly model the operation of pumps. When a pump is off, the result is a disconnection of a source from the network. In that case, (5) will require the inclusion of integer variables. When using variable speed pumps, modelling such a disconnection could be achieved without the use of integer variables. A pump will be considered off when pump flow is equal to zero. The pump rotation speed variable will then act as a free variable to adjust the head at the pump's downstream node, virtually allowing the pump to function as a valve and regulate a virtual head gain along the pump.

The inclusion of variable speed pumps allows formulating the problem as a continuous non-linear problem and helps avoid the employment of binary variables, although the inclusion of regular pumps (i.e., single speed pumps) would significantly reduce the solution space on one hand, while limiting the flexibility of the WDS design and operation on the other hand, which is essential for presenting the ability to adjust the WDS operation to the operation of the PG in this particular work.

On the PG side, the problem includes the following constraints,

which constitute the relationship between power flow and voltage drop along transmission lines, bus power balance, generation limits, and voltage magnitude bounds:

$$Y_m = \frac{1}{2} (e_m e_m^T Y + Y^H e_m e_m^T) \quad \forall m \in M \quad (10)$$

$$\bar{Y}_m = \frac{j}{2} (e_m e_m^T Y - Y^H e_m e_m^T) \quad \forall m \in M \quad (11)$$

$$p_m^t = (v^t)^H \bullet Y_m \bullet v^t \quad \forall m \in M \forall t \in \mathcal{T} \quad (12)$$

$$q_m^t = (v^t)^H \bullet \bar{Y}_m \bullet v^t \quad \forall m \in M \forall t \in \mathcal{T} \quad (13)$$

$$p_m^t = p_{in,m}^t - p_{L,m}^t - \frac{1}{FOC} \sum_{l \in L} \left(I_{m,l} \bullet \frac{\gamma H_{p,l}^t Q_{p,l}^t}{\eta_p} \right) \quad (14)$$

$$q_m^t = q_{in,m}^t - q_{L,m}^t - \frac{1}{FOC} \sum_{l \in L} \left(I_{m,l} \bullet \alpha_{qp} \frac{\gamma H_{p,l}^t Q_{p,l}^t}{\eta_p} \right) \quad (15)$$

$$p_{in,k} \leq p_{in,k}^t \leq \bar{p}_{in,k} \quad \forall t \in \mathcal{T} \quad (16)$$

$$\bar{p}_{in,k} = SF_k \bullet SGC \quad \forall k \in m^* \quad (17)$$

$$\left(\underline{v} \right)^2 \leq (v^t)^H \bullet M_m \bullet v^t \leq (\bar{v})^2 \quad \forall m \in M \forall t \in \mathcal{T} \quad (18)$$

$$v_1^t = v_{ref} \quad \forall t \in \mathcal{T} \quad (19)$$

where: Y is the power grid's admittance matrix; e_m is a standard basis vector containing 1 in the m^{th} element; m is the bus index; M is set of buses in the grid; t is the timestep; $(\bullet)^H$ is the complex-conjugate transpose; $p_{in,m}^t$ and $q_{in,m}^t$ are the active and reactive power generated at bus m , respectively; $p_{L,m}^t$ and $q_{L,m}^t$ are the active and reactive loads, respectively; $I_{m,l}$ is an indicator parameter indicating which buses are connected to pumps in the grid; γ is water specific weight [N/m³]; $H_{p,l}^t$ is the pump head gain [m]; $Q_{p,l}^t$ is the pump flow [m³/sec]; α_{qp} is the pump's active and reactive ratio; η_p is the pump efficiency; FOC is the factor of conversion, converting pump power from metric units into the p.u. unit system, set here as 1 [MW/p.u.]; m^* is the index corresponding to the bus where solar power generators are installed; $p_{in,k}$ and $\bar{p}_{in,k}$ are the lower and upper generation bounds for generator k , respectively; SGC is the solar generation curve, presented in Fig. 1; $\underline{v}$ and $\bar{v}$ are the lower and upper voltage bounds, respectively; M_m is a matrix defined by the standard basis vectors, such that $M_m = e_m \bullet e_m^T$, and v_{ref} is the voltage value defined for the reference bus.

2.3. Objective function

Usually, the objective function of optimal WDS design problems is defined as the minimal cost of both operation and construction derived from the design. In this paper, the results of two different problems will be presented and compared: the first being an optimal design problem of an independent WDS, with its results imposed on the power grid as a given load; and the second being the conjunctive optimal design problem, when both systems are designed simultaneously. Here, both the objective function of the independent WDS design problem and the objective function of the conjunctive design problem will be presented. When formulating the independent design problem, energy tariffs were not considered for the operation of pumps. Those tariffs are usually the result of an optimization done by the PG operators. It was the authors' goal to present a reasonable comparison between the solutions to the two design problems, and therefore the decision was made not to consider energy tariffs. This matter is addressed in more detail in the conclusion section. The objective function of the independent problem comprises of the cost of constructing all pumping stations and tanks in the system. Those costs can be calculated by the following expressions:

$$PUCC = \frac{CPUMP}{270 \cdot \eta_p} \sum_{l=1}^{npumps} \max(H_{p,l}^1 \cdot Q_{p,l}^1, \dots, H_{p,l}^T \cdot Q_{p,l}^T) \quad (20)$$

$$TCC = \sum_{s=1}^{ntanks} UTC_s \cdot \max(H_{tanks,s}^1, \dots, H_{tanks,s}^T) \quad (21)$$

where $PUCC$ is pump construction cost [MU]; $CPUMP$ is unit pump cost of pump construction [MU/hp]; TCC is tank construction cost [MU]; UTC_s is unit water level cost of tank s [MU/m].

And the objective function for the independent design problem is given in (22):

$$Z_1 = \min TCC + PUCC \quad (22)$$

For the conjunctive design problem, two cost components are added to the objective – the cost of generating power, and the cost of constructing a solar field. Those are given by the following expressions:

$$PGC = \left(\sum_{i \in \mathcal{I}} \sum_{k \in K} a_k (p_{in,k}^i)^2 + b_k (p_{in,k}^i) + c_k \right) \cdot AD \cdot APGPV \quad (23)$$

$$SFC = w \cdot \left(\sum_{k=1}^{nsolargenerators} SF_k \right) \quad (24)$$

where PGC is power generation cost [MU]; a_k [MU/p.u.²] b_k [MU/p.u.] and c_k [MU] are generation cost coefficients of generator k ; AD is annual duration [days]; $APGPV$ is annual power generation present value coefficient; $SFCC$ is solar field construction cost [MU]; and w is the solar field construction cost coefficient [MU/MW].

And the objective function for the conjunctive design problem is given in (25):

$$Z_2 = \min TCC + PUCC + PGC + SFC \quad (25)$$

Table 1 details the cost parameters that were used. Where WDS cost parameters are based on Ostfeld (2005).

3. Example applications

The conjunctive optimal design problem was applied here to two illustrative examples – one relatively simple, and another slightly more complex – for the purpose of testing the effectiveness of the approach on different scales. In both examples, the same water and power demand patterns presented in Fig. 1a, along with the solar generation profile in Fig. 1b, were used. Tanks were assumed to have a constant cross-section

Table 1

Cost parameters used for objective functions in the optimal design problem (Ostfeld 2005).

Parameter	Value
a_k [MU]	5
b_k [MU]	50
c_k [MU]	10
AI [%]	5
$TPLD$	40
$APPV$	10.04
$CPUMP$	1,000
UTC	20,000
AD [days]	365
$APGPV$ [-]	0.9524
w [MU/W]	0.08

MU – money units.

area, and therefore the volume decision variable was replaced with the tank's maximal level. An initial condition was set for the tanks to be full at the beginning of the system's simulation. The model was coded using YALMIP optimization package through MATLAB, which is a MATLAB package that allows the straightforward formulation of optimization problems (YALMIP, 2023). Although the problem is quite complex, the examples presented below did not require any intervention in the solution process of the development of a solution strategy, and they were simply solved using IPOPT local solver, with a 64-bit Intel i7 4-core CPU at 2.00 GHz and 8-GB RAM. The optimization horizon was set to equal 24 h. However, since the problem is highly complex, timestep length was set to 3 h, resulting in a solution for 8 timesteps.

3.1. Illustrative example I

Example I comprise of a four-node WDS, with one tank located at node 4 and one pump, which is supplied with power by a three-bus power grid that contains one conventional generator that is located at bus 1. The system layout is presented in Fig. 2.

The WDS and power grid attributes are given below in Table 2.

3.1.1. Results

The results of the base run are reported in Table 3, and they are compared to the results of the optimal design of the independent WDS.

Combining the two systems under one optimization model has an impact. The results show a difference in tank volume, with a slight difference in pumping capacity between the independent optimal design and the conjunctive optimal design. A drop in total cost is observed in

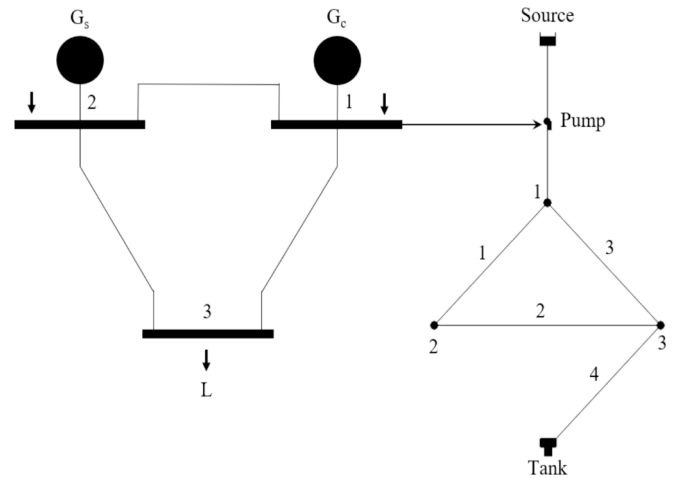


Fig. 2. Example I – system layout (left – power grid with a conventional generator at bus 1, a solar generator at bus 2, and an external load at bus 3; right – WDS).

Table 2

Example I – system attributes.

System	Component	Attribute	Value
Water distribution system	Link	Diameter [mm]	400
		Roughness [-]	110
		Length [km]	3
	Node	Base water demand [m ³ /hr]	800
		Pressure demand [m]	30
		Elevation [m]	0
	Tank	Elevation [m]	30
		Diameter [m]	36
	Tank link	Diameter [mm]	500
		Length [km]	0.5
	Pump	η [-]	0.85
		Curve coefficients	200; 1×10^{-5}
		Rotation speed range [rpm]	[0,50]
		α_{pq} [-]	$\frac{1}{3}$
Power grid	Transmission line	Impedance [p.u.]	0.1j
	Capacitor	Admittance [p.u.]	0.008j
	Bus	Minimum voltage bound [p.u.]	0.95
		Maximum voltage bound [p.u.]	1.05
	Reference bus	Reference voltage [p.u.]	1
	Load	Base power demand [p.u./hr]	2.8653 + 1.2244j

Table 3

Example I – independent and conjunctive design results comparison.

	Independent	Conjunctive
Tank volume [m ³]	9,706	9,281
Pumping station capacity [kW]	359	366
Solar field capacity [MW]	–	3.3
Pumping station construction cost [MU]	488,323 (31 %)	497,460 (35 %)
Tank construction cost [MU]	190,721 (12 %)	182,370 (13 %)
Power generation cost [MU]	919,215 (58 %)	493,471 (34 %)
Solar field construction cost [MU]	–	265,995 (18 %)
Total cost [MU]	1,598,260	1,439,296

Parenthesized – percent out of total cost.

the conjunctive approach compared to the independent one, with the most prominent reduced cost component being power generation cost. Evidently, allowing the construction of a solar generation field leads to a reduction in conventional generation, which translates directly into a reduction in generation cost, but is transformed into an investment in solar field construction. Nonetheless, the sum of solar field construction and conventional generation costs is still less than power generation costs in the independent design. The sustainable energy source

encourages higher pumping capacity, which in result leads to a smaller tank size. This can be interpreted as a direct link between the type of power source to the optimal capacities of the different facilities inside the water system.

Fig. 3 presents the WDS optimal operation policy derived from the obtained optimal design, including pump power demand (Fig. 3a) and tank level (Fig. 3b).

The operational policies obtained in each of the considered problems seem quite similar (Fig. 3a and 3b). The larger tank size obtained in the independent problem helps to slightly reduce pump power demand during the morning and noon, when both power and water demands peak. The tank volume is not fully utilized as a result of both the nodal head constraint, which is binding at 17:00 in node 2, and the closure constraint in Equation (3), which forces the pump to operate in higher capacity to fill it and makes it less attractive to empty the tank.

Fig. 4 compares the PG operation policies obtained by the independent and conjunctive optimal design problems. Depicted are the power generation quantities (Fig. 4a and 4c) and voltage amplitudes at all buses of the PG (Fig. 4b and 4d).

It can be observed in Fig. 4c that although the solar capacity is chosen to be 3.3 MW, the actual solar generation does not exceed 2 MW. As mentioned, the solar capacity variable is multiplied by the solar generation curve presented in Fig. 1b to create the solar generation limit. Therefore, constructing a 3.3 MW capacity solar field will allow the system to generate enough solar power required during off-peak hours, such as morning and evening.

As can be observed in Fig. 4b the independent optimal design produces an infeasible solution from the standpoint of the power grid, as minimal voltage bound constraints cannot be satisfied. This result highlights the importance of the conjunctive approach and advocates for the consideration of intertwined systems when designing a water distribution system.

3.1.2. Sensitivity analysis

To test the model and its validity, a sensitivity analysis comprising of 7 scenarios was conducted. This analysis was applied only to the conjunctive problem. In each of the scenarios, a slight modification was made to one of the parameters in the base run, as presented in Table 4. The scenarios were designed to highlight different aspects of the problem and they way that they would influence it. The magnitudes of the modifications were set randomly.

The results of the sensitivity analysis are presented in Fig. 5, and include the tank volume (5a), pumping capacity (5b), solar capacity (5c) and objective value (5d).

Widening the pipe diameters in scenario 1, demonstrating the effect of including pipe diameters in the optimization, is expected to lead to reduced power demand in the WDS, as the energy dissipation in pipes is reduced. We can observe that pumping capacity in fact reduces, along with tank volume, shrinking to 3,940, which is less than half the value

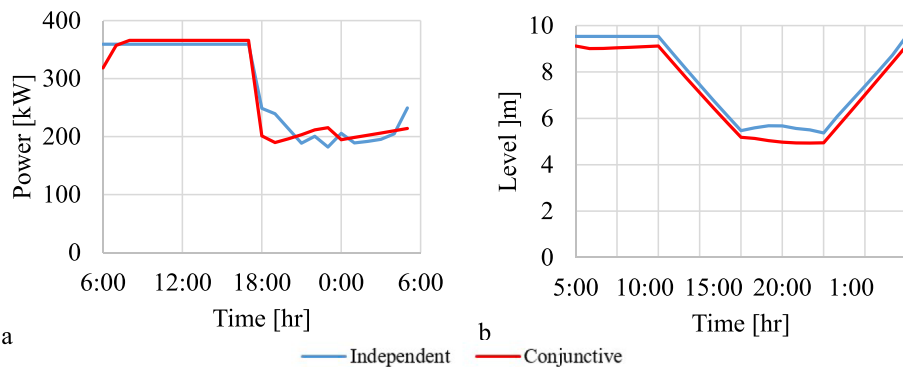


Fig. 3. Example I – comparison of operation results obtained for pump (a) and tank (b) by independent and conjunctive optimal design.

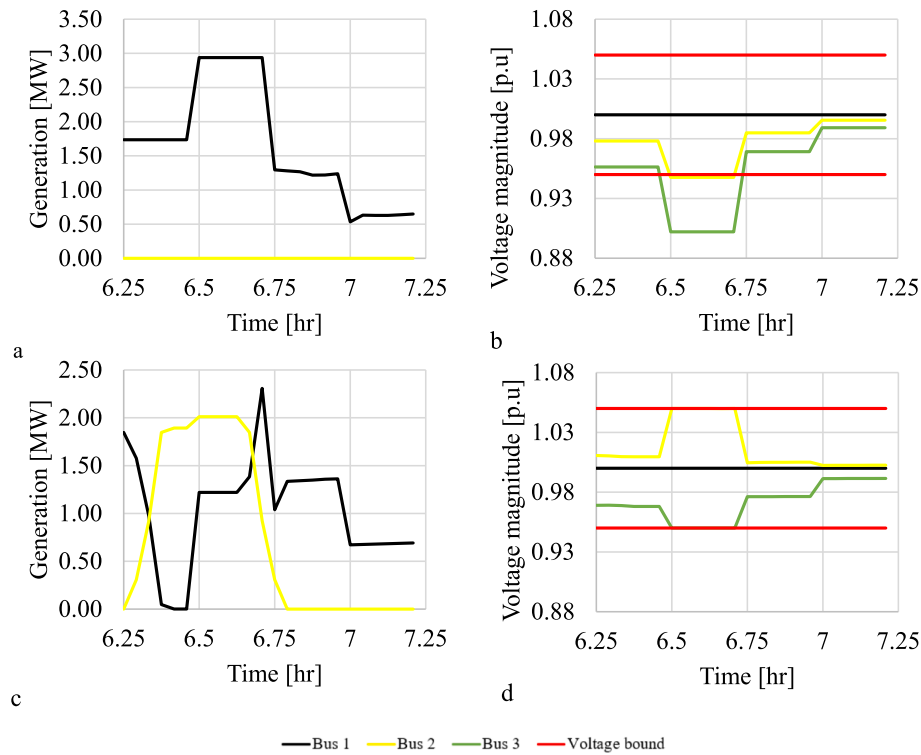


Fig. 4. Example I results – (a) independent generation (b) independent voltage magnitudes (c) conjunctive generation (d) conjunctive voltage magnitudes.

Table 4

Example I – sensitivity analysis scenarios (cells highlighted in grey contain the modified parameter in each scenario).

Scenario	1	2	3	4	5	6	7
Pipe diameters [mm]	600	400	400	400	400	400	400
Water base demand [m ³ /hr]	800	960	800	800	800	800	800
Power base demand [MW]	2.8653 + 1.2244j	2.8653 + 1.2244j	2.2922 + 0.9795j	2.8653 + 1.2244j	2.8653 + 1.2244j	2.8653 + 1.2244j	2.8653 + 1.2244j
Line impedance [p.u.]	0.1j	0.1j	0.1j	0.05j	0.1j	0.1j	0.1j
Number of consumers	3	3	3	3	4	3	3
Service pressure [m]	30	30	30	30	30	20	30
Solar field construction cost [MU/W]	0.08	0.08	0.08	0.08	0.08	0.08	0.004

obtained in the base run, and the smallest value out of all examined scenarios. Due to the system being of a small scale, the tank operates in tight coordination with the pump, and as pump head gain is reduced, so does tank level – being the two components that govern the distribution of energy in the system. As the conjunctive design problem is already quite complex, pipe diameters were omitted as decision variables, and it was the authors' decision to concentrate on energy facilities in the WDS and PG. However, it should be mentioned here that the result of scenario 1 demonstrate the importance of including diameters in the design problem. Modifying water base demand, demonstrating a scenario of increased water demand by a factor of 1.2 due to population growth, results in the highest tank volume and pumping capacity out of all scenarios. To deal with the increase in water demand, a higher capacity pumping station must be constructed, along with a larger tank, to balance energy costs. No change is observed in solar capacity, leading to the conclusion that it is more expensive to increase the solar capacity than to conventionally generate the difference in power. This result means that a population growth that would lead to increased water demand is likely to not be translated in larger renewable energy capacity in the PG. Reducing base power demand in scenario 3 results in a slight reduction

of tank volume and a slight rise in pump capacity. The most prominent effect of this modification is a reduction of approximately 700 kW in solar capacity and a drop in objective value. A similar trend is observed after reducing line impedance to 0.05j, to demonstrate the effect of including transmission line material in the optimization (equivalent to WDS pipe diameters), although in this case the drop in solar capacity is less prominent. Reducing the line impedance results in less power being lost along transmission lines, and in turn allows to generate less power to satisfy the demand. Reducing the strain on the power grid allows for larger quantities of power to flow to the pump, and a reduction in tank volume is observed as a result. This result emphasizes that an improved PG could significantly benefit the operation of the WDS and save costs in design. In scenario 5, a fourth water consumer is connected to the WDS by closing the loop 1–3–4, to demonstrate an expansion of the WDS. The total water consumption is kept as is, but is now equally split between the four consumers. A reduction in both tank size and pumping capacity is observed, with no change in solar capacity. A likely explanation for this result is that due to the additional pipes in the system, the distribution of flow leads to reduced longitudinal head loss, in a similar logic to the result of scenario 1. This result further advocates for the inclusion

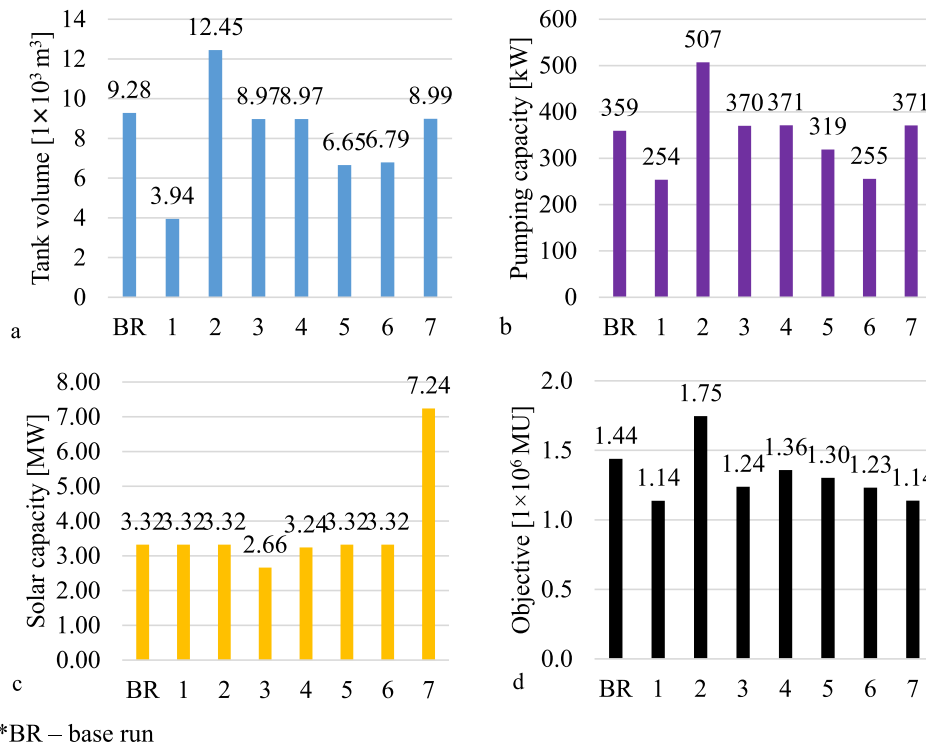


Fig. 5. Example I – sensitivity analysis results. *BR – base run.

of pipe diameters as part of the optimization problem. Lowering service pressure to 20 m, to demonstrate a relaxation of pressure constraints, as done in scenario 6, seems to lead to the same trend – a reduction in tank size and pumping capacity, with no significant change in solar capacity. This result demonstrates the dominant influence that pressure constraints impose on the design and shows that results could significantly change based on the system's scale and designation. The three scenarios addressing energy parameters in the WDS – scenarios 1, 5 and 6, all result in the lowest values of tank size and pumping power, with the same value of solar capacity as obtained in the base run. In scenario 7, reducing the cost of solar field construction to 0.004 MU per W to demonstrate the availability of a more efficient renewable energy source results in a steep rise in solar capacity, and a rise in pumping capacity as a result. We now obtain much higher tank size and pumping capacity compared to scenario 1, in which the objective value is the same. This result indicates how beneficial the development of cheaper and more efficient power sources can be, and their potential to improve WDS design.

3.2. Illustrative example II

Example II is slightly more complex compared to Example I. It comprises of a nine-node WDS with 11 pipes, two tanks located in nodes 8 and 9, and two pumps placed on links 1 and 2. A 14-bus PG supplies the pumps with power, along with 6 other demand buses along the PG. Similar to example I, the PG includes one conventional generator located at bus 1, and a solar generation field to be constructed and connected to bus 2. The system layout is presented in Fig. 6. The existence of multiple pumping stations and multiple tanks allows higher flexibility in the design and operation, which significantly enlarges the solution space and complexifies the problem. Furthermore, this would allow the examination of trade-offs between different zones inside the WDS, and will enable to observe how each of them is affected by the power grid. On the PG side, more power consumers and transmission lines result in a more complex optimal power flow problem, which would also influence the complexity of the conjunctive problem and help gain meaningful

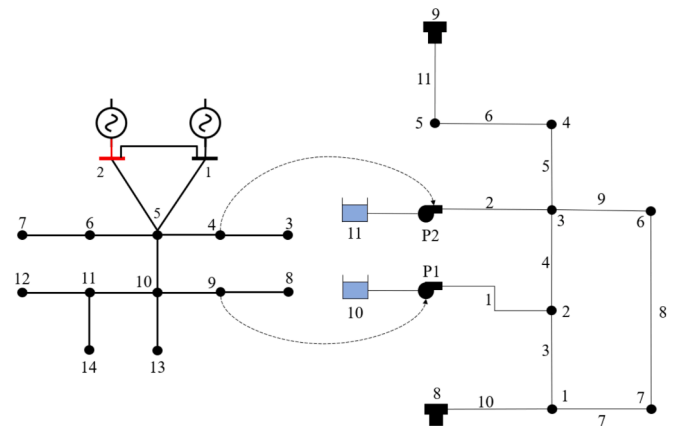


Fig. 6. Example II – system layout (left – 14 bus power grid with a conventional generator at bus 1, solar generator at bus 2, and power consumers at buses 3, 7, 8, 12, 13, 14; right – WDS).

insights into the design of the joint system.

The attributes of the WDS and the PG are given in Table 5.

In this example, the 24-hour optimization was divided into 3-hour long timesteps (meaning – a total of 8 timesteps), due to the complexity and difficulty of achieving a solution for a 24-timestep long optimization. Since the focus of this study is the optimal design of the two systems, the operational decisions can be grossly estimated with longer time steps that relax the computational challenge.

3.2.1. Results

The solution of the conjunctive optimal design problem is brought in Table 6, alongside a comparison to the solution of the independent optimal design problem.

The two tanks and two pumps in the WDS seem to respond differently to the connection of the PG to the optimization problem. Compared to the results of the independent design problem, the size of tank 8 is

Table 5

Example II – system attributes.

System	Component	Attribute	Value
Water distribution system	All links	Diameter [mm]	300
		Roughness [-]	120
	Links 3,4,5,6,7,9	Length [km]	2
	Link 8		4
	All nodes	Pressure demand [m]	30
		Elevation [m]	0
	Nodes 4,5	Base water demand	150
	Node 6	[m ³ /hr]	300
	Node 7		400
	Tanks	Diameter [m]	20
		H _{min} [m]	40
	Tank links (10, 11)	Diameter [mm]	400
		Length [km]	0.4
	Pumps	η [-]	0.85
		Curve coefficients	200; 1×10^{-5}
Power grid		Rotation speed range [rpm]	[0,50]
		α_{pq} [-]	$\frac{1}{3}$
	Transmission line	Impedance [p.u.]	0.006 + 0.01j
	Bus	Voltage magnitude [p.u.]	1
	Reference bus	Minimum voltage bound [p.u.]	0.95
		Maximum voltage bound [p.u.]	1.05
	Loads (buses 3,7,8,12,13,14)	Base power demand [p.u./hr]	0.5 + 0.35j

Table 6

Example II – independent and conjunctive design results comparison.

	Independent	Conjunctive
Tank 8 vol [m ³]	1,514	1,402
Tank 9 vol [m ³]	300	320
Pumping station 1 capacity [kW]	69.6	46.7
Pumping station 2 capacity [kW]	159.3	185.5
Solar field capacity [MW]	–	2.09
Pumping stations construction cost [MU]	311,070 (43 %)	315,471 (41 %)
Tank construction cost [MU]	115,360 (16 %)	108,792 (14 %)
Power generation cost [MU]	292,971 (41 %)	176,089 (23 %)
Solar field construction cost [MU]	–	166,833 (22 %)
Total cost [MU]	719,402	767,185

Parenthesized – percent out of total cost.

reduced, while the size of tank 9 is slightly increased; the capacity of pump 1 is reduced, and the capacity of pump 2 is increased. This result reflects the effect of the system's structure – as water demand is larger in the top area of the WDS, it would be more beneficial to invest in the construction of pump 2 and tank 9, which are located more adjacent to the center of demand. Another factor is the power grid itself – pump 1 is connected to bus 9, which lies further away from the generators in the power grid. Therefore, the voltage drop between the generators and bus 9 is more significant, which makes it less attractive to supply power to pump 1, and encourages the system to rely on pump 2, which is more adjacent to power generation components. This analysis is supported by Figs. 7 and 8, which respectively show the WDS and PG operational policies obtained by the design, and compare the operation suggested by the independent solution and the conjunctive solution.

The conjunctive solution seems to allow a more efficient operation of tank 9, emptying it completely twice a day (Fig. 7d). Although we can observe that pump 2 is more dominant in the operation and operates at a higher capacity (Fig. 7b), pump 2 is turned off during the night hours and pump 1 is responsible for filling the tanks to meet their closure constraint at the end of their operational cycle (Fig. 7a).

The same phenomenon that was presented in example I occurs here –

the optimal design obtained for the WDS is not feasible from the PG standpoint, and a solar generator must be constructed to satisfy voltage magnitude constraints (Fig. 8b and 8d). In this case, the conjunctive solution produces a more expensive solution compared to the solution of the independent problem. This result stems from the combination of the dominance of solar power generation inside the power grid and the voltage boundaries. In this case study, solar generation constitutes a smaller percentage out of the total power demand in the PG, and while it does reduce power generation costs quite dramatically, it is not able to reduce them to the point where the total cost is reduced below the independent solution total cost. Although not constructing a solar field at all is deemed a cheaper solution, voltage constraints cannot be satisfied this way, which forces its construction and creates a more expensive solution, which satisfies all constraints.

In the independent solution, no generator is connected to bus 2 and therefore there is no power generation there (Fig. 8a). As opposed to the results of example I, now we can observe that the solar capacity is exploited to its fullest, reaching the maximal value of 2.09 MW at 12:00 (Fig. 8c).

3.2.2. Sensitivity analysis

A more limited sensitivity analysis was performed to gain a slightly deeper insight into the interaction between the different components in this system. The examined scenarios are detailed in Table 7.

The results of the sensitivity analysis are shown in Fig. 9. *BR – base run.

Scenarios 1 and 2 examined the solution's sensitivity to solar field construction cost. When solar field construction cost is reduced, we do not observe a significant change in results. When the cost increases, however, we see a reduction in the size of tank 8; an increase in size of tank 9; a reduction in pumping capacity of pump 1; and an increase in capacity of pump 2. This result validates the analysis that was previously presented, about the effects the different components have on the operation, and their relationship with the availability of sustainable power. We can also notice a 70-kW reduction in solar capacity, as a result of increased construction cost. The third scenario examines a different configuration of the system, where pump 2 is connected directly to bus 2, where the solar field is located. We obtained the same results as we did in scenario 2 but are now able to operate with a much smaller solar field. This result demonstrates the magnitude of the influence that both the voltage constraints and the loss of power along transmission lines have on the optimal solution.

4. Conclusion

This paper presents the novel concept of conjunctively optimizing the design of a water distribution system and a power grid. The optimization problem is formulated such that the decision variables are storage and pumping capacities on the WDS side, and solar generation capacity on the power grid side. The conjunctive problem comprises of all constraints forming each of the two independent design problems, and the two systems are linked through the WDS's pump power demand.

The concept is applied to two illustrative examples. The results of the conjunctive optimal design problems are compared to the results of the optimal design of the independent WDS, and the impact of the connection of the power grid to the WDS optimal design problem is noticeable. The inclusion of voltage constraints in the problem reveals the inability of the power grid to operate under the independent WDS optimal design requirements and forces the connection of the grid to a sustainable power source. The availability of a sustainable power source in the system influences the design of storage and pumping components in the WDS, favoring the upgrading of pumping stations at the expense of storage volume – a somewhat counterintuitive result that advocates for the importance of the conjunctive approach. The examination of two illustrative examples casts light on the complexity of the problem and the inner mechanisms that are at play. Analyzing the results of example I

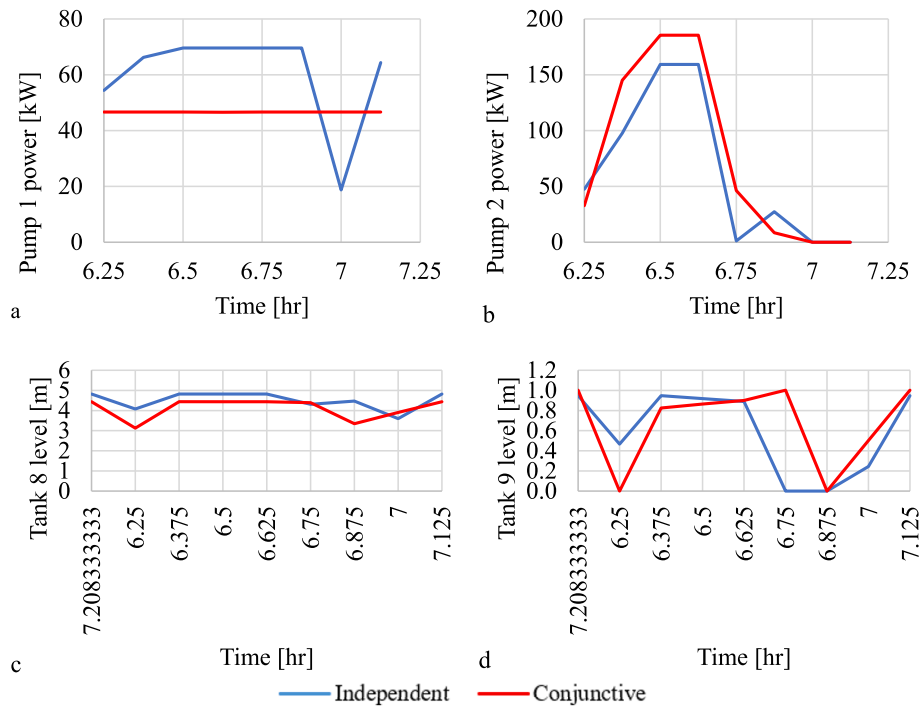


Fig. 7. Example II – comparison of operation results obtained for pumps (a,b) and tanks (c,d) by independent and conjunctive designs.

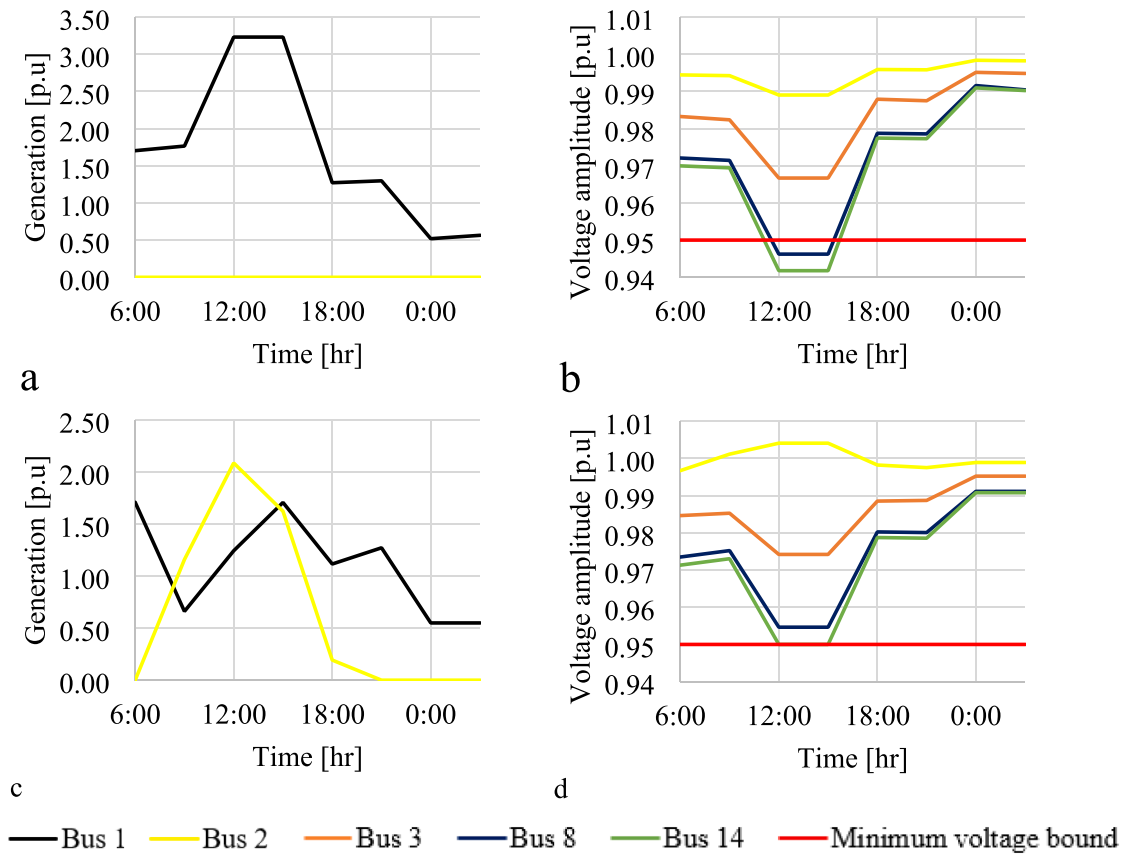
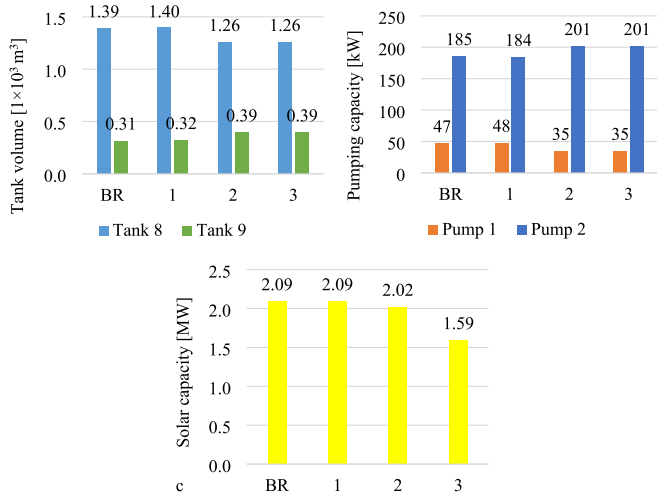


Fig. 8. Example II – (a) independent generation (b) independent voltage magnitudes (c) conjunctive generation (d) conjunctive voltage magnitudes.

Table 7

Example II – sensitivity analysis scenarios.

Scenario	1	2	3
Solar field construction cost coefficient [MU/W]	0.06	0.10	0.08
Pump 2 bus connection	4	4	2

**Fig. 9.** Example II – sensitivity analysis results.

could be considered a simple task, since the system includes only one tank and one pump, but as the system is of a slightly larger scale the relationship between components can become quite complex, even when looking at a relatively small system such as the one brought as example II. The different components respond differently to the connection of a sustainable source to the system, based on their adjacency to demand nodes and their connections to the power grid. The PG configuration favors the employment pumps located in the proximity of generation epicenters. Storage facility volumes inside the WDS are in turn affected by this based on their proximity to pumps, as they must operate in coordination with pump head gains. Regarding the cost, no clear trend was observed, as the conjunctive approach is able to reduce cost for example I but is not able to do the same for example II. As power generation is the most dominant component in the objective, it is evident that the dominance of the renewable energy source inside the power grid is responsible for the ability to reduce the total cost. Approximating the dominance of a renewable energy source in the PG is therefore a step that should be taken before the application of this conjunctive model, in order to assess whether total cost could be reduced.

A sensitivity analysis that was performed helped establish and validate the obtained results, and to also gain novel insights into the optimality of the obtained solution. It was observed that pipe diameters, along with transmission lines, could be very significant decision variables in this problem, and their inclusion could provide more deep and interesting insight.

An issue that required the authors' attention is the basis for comparison between the independent design and the conjunctive design. Usually, the operation of WDS takes into consideration energy tariffs for pumps. Those tariffs are imposed on the WDS by the PG operators. Since the design and operation of the PG are also being solved as part of the conjunctive approach, the incorporation of those energy tariffs in the conjunctive problem would defeat the purpose of this work – here, the model is intended to be solved all at once, simultaneously, rather than in stages (optimizing the design and operation of the PG, obtaining energy tariffs for the WDS, and then optimizing the operation of the WDS with respect to the obtained tariffs), in order to utilize the interdependency between the systems.. In this work, we resolved this matter by assuming

that a single operator oversees the operation of both the WDS and the PG. This assumption allows to neglect pump energy tariffs and only consider power generation costs from the standpoint of the PG. Although it is rarely the case in reality, it does allow us to present a relatively beneficial comparison in this work. Another option could be performing a calculation of energy tariffs in a dynamic way, during the optimization. This option is much more complex. It requires a focus on the methods that are used for the determination of energy tariffs by PG operators and regarded by the authors as out of the scope of this paper. That said, the incorporation of energy tariffs does require more attention and poses a significant challenge that could be the focus for future work.

The optimization problem presented here is solved by IPOPT, an off-the-shelf non-linear solver. As the networks get more complex, the process of obtaining a solution this way would get more difficult, and at some point, perhaps impractical. Nonetheless, applying this conjunctive approach to case studies of a larger scale will help better understand the magnitude of their interdependency on optimal design of such systems, and is imperative for advancing the field of infrastructure operation and management. An additional work that's currently being carried out is the development of a tailored solution method, that would allow the application of the conjunctive approach to larger case studies, and will shed light on new insights with respect to the scale of the problem.

As mentioned throughout this manuscript, the work presented here constitutes a first step in the direction of a more holistic approach for water distribution systems analysis, specifically with a focus on their relationship with the power grid. There are several suggestions that could be of use for future work that would expand the idea that was brought here. Widening the range of decision variables, to include pipe diameters, transmission line materials and even batteries for power storage in the power grid side would significantly complexify the problem, and is likely to generate rich insight into the interdependencies between the systems. The addition of decision variables is also likely to require more sophisticated and computationally complex solution approach, such as branch and bound, decomposition techniques like outer approximation (OA), heuristics or sequential quadratic programming (SQP). Testing the model on a much larger scale systems is an interesting direction as well, which will require a closer look at the problem formulation in order to develop new and tailored solution methods. The development of such methods could be another research direction in and of itself, to allow for a more suitable optimization approach and provide robustness to the optimization process. In pursuing any of these directions, different objectives could be considered such as robustness, resilience or even water quality standards. A multi-objective problem could be tackled, and the competition between those different objective functions could be presented as a pareto front, to demonstrate the limitations of the conjunctive approach. Different aspects of uncertainty could also be examined, using different methods to solve the operation part of the problem in a more realistic way, such as robust optimization, folding horizon and predictive control models.

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CRediT authorship contribution statement

Tomer Shmaya: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Avi Ostfeld:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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